Benchmark notes for `lebedev_em`

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Abstract

These notes document the benchmarks for the `lebedev_em` solver, a Python implementation of the Lebedev staggered-grid finite-difference scheme of Davydycheva, Druskin & Habashy [1] (DDH03) for the frequency-domain Maxwell equations in 3-D anisotropic inhomogeneous media. Two inhomogeneous benchmarks are presented. The first, a two-half-space model with a 10× conductivity contrast, is validated against the exact Sommerfeld-integral solution: the four-cluster Lebedev average attains ≈ 1% accuracy in the conductive half-space, and cluster averaging reduces the single-cluster error fivefold. The second reproduces the configuration of DDH03 Figure 9 — a resistive borehole with an invaded zone crossing a thin 75° dipping anisotropic layer with $\sigma_N = \sigma_T/200$ — and matches the published curves at the few-percent level with unit dipole moment and no fitted parameters, *identically at every grid resolution*, using either the DDH03 eq.-(8/9) laminate (Backus) averaging or the nodal homogenization of [2] in a fully coupled four-cluster solve. A three-level refinement study confirms the resolution-independence that is the central claim of the DDH03 sub-cell averaging approach: the two averaged schemes agree with each other to ≈ 1% and with the published curves equally well at every grid level.

1 The solver and its verification chain

`lebedev_em` discretizes the frequency-domain Maxwell system (time convention $e^{-i\omega t}$)

$$\nabla \times \mathbf{E} = i\omega\mu\mathbf{H}, \qquad \nabla \times \mathbf{H} = \dot{\sigma}\,\mathbf{E} + \mathbf{J}, \qquad \dot{\sigma} = \sigma - i\omega\varepsilon, \tag{1}$$

on Lebedev's staggered grid, which supports full 3×3 conductivity tensors without interpolation and splits into four Yee-like clusters. The implementation follows the complete DDH03 methodology: distributed sources in all four clusters satisfying their eq. (7) (unit total weight, centroid at the physical source), mixed electric/magnetic boundary conditions assigned per cluster so that domain-truncation errors cancel in the four-cluster average, interpolate-then-average field extraction, optimal geometric transverse grids ($\alpha = e^{\gamma\pi/\sqrt{k}}$, $\gamma = 1/\sqrt{2}$) with a hybrid axial grid (fine equidistant near source and receivers, geometric beyond), and sub-cell homogenization of interface-cut cells.

The verification chain proceeds from analytic anchors outward: the discrete operators are unit-tested (symmetry of the curl–curl system in its volume-weighted inner product, source weight and centroid conditions on nonuniform grids); the solver reproduces the closed-form magnetic-dipole solution in homogeneous isotropic media at both 2.5 kHz and 52.65 kHz to a few percent at moderate resolution; and the two inhomogeneous benchmarks below add a sharp material interface and a thin anisotropic structure. All results are per unit dipole moment (1 A·m^2) in SI units, with no calibration constants anywhere in the chain.

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2 Benchmark 1: two half-spaces, $10\times$ contrast

Model. A unit vertical electric dipole (VED) at the origin in

$$\sigma(z) = \begin{cases} 0.1 \text{ S/m}, & z < 4 \text{ m}, \\ 1.0 \text{ S/m}, & z \geq 4 \text{ m}, \end{cases} \qquad f = 2500 \text{ Hz},$$

with receivers on the z-axis, $z \in [0.5, 7.5]$ m, straddling the contact. The reference is the exact Sommerfeld-integral solution for E_z (implemented in `analytics.py` and independently verified: it reduces to the closed-form homogeneous solution when $\sigma_1 = \sigma_2$ to 3×10^{-14} relative, and satisfies the normal-current continuity $\sigma_1 E_z^- = \sigma_2 E_z^+$ at the contact).

Grid. Optimal geometric transverse grids with $k = 3$ ($M_x = M_y = 12$), hybrid axial grid with $\Delta z = 0.0625$ m near the source ($M_z = 160$, ≈ 41k complex degrees of freedom).

Results. Figure 1 shows $|E_z|$ on the axis, the pointwise deviation from the analytic solution, and the contact effect.

- In the conductive half-space ($z \geq 4$ m), across the $10\times$ jump, the RMS relative error of $\operatorname{Re} E_z$ is **1.0**%.
- The overall RMS error (5.6%) is dominated by the near-source region, where the transverse domain is only $0.3\,\delta_1$ wide; the *same* error appears in a homogeneous control run (RMS 5.7%), so the interface itself contributes essentially nothing — the scheme crosses the contact cleanly.
- The physical structure at the contact is captured quantitatively: the two-layer/homogeneous ratio rises to 1.65 just above the contact and drops to ≈ 0.18 below it, matching the analytic curve point by point.
- Cluster averaging matters: a single cluster (C000) gives RMS 29.9%; the four-cluster Lebedev average reduces this to 5.7% — DDH03's error-cancellation mechanism at work in an inhomogeneous medium.

3 Benchmark 2: thin dipping anisotropic layer crossed by a borehole

Model. The configuration of DDH03 Figure 9 (Fig. 2): a homogeneous background $\sigma = 0.1$ S/m; a resistive borehole column ($\sigma = 0.05$ S/m, $R = 0.1$ m) along the z-axis; an invaded zone ($\sigma = 0.1$ S/m, $R = 0.6$ m) that replaces the formation near the wellbore; and a thin dipping anisotropic layer of 0.25 m normal thickness, dip 75°, with transverse conductivity $\sigma_T = 0.1$ S/m and normal conductivity $\sigma_N = \sigma_T/200$ — a 1:200 contrast confined to a layer thinner than most local grid cells, present for $r \geq 0.6$ m. The layer crosses the borehole axis over $z \in [-1.93, -0.95]$ m. The source is a z-directed unit magnetic dipole at the origin, $f = 52.65$ kHz; the observable is $\operatorname{Im} B_z$ on the axis. Two cases are run: *no layer* (borehole only) and *resistive layer*.

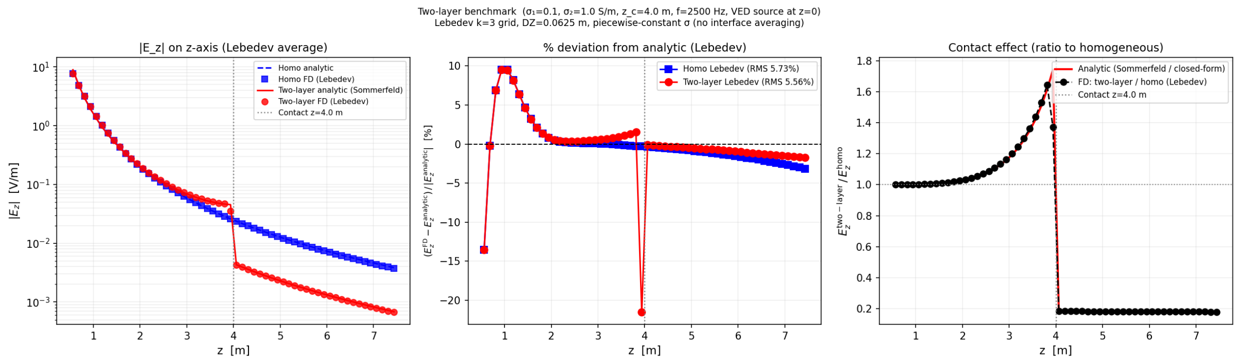

Figure 1: Two-half-space benchmark ($\sigma_1 = 0.1$, $\sigma_2 = 1.0$ S/m, contact at $z_c = 4$ m, $f = 2500$ Hz). Left: $|E_z|$ on the axis, FD (symbols) vs. analytic (lines). Center: pointwise percent deviation. Right: contact effect — two-layer to homogeneous ratio, FD (black) vs. Sommerfeld analytic (red). The isolated spike in the center panel sits at the contact node itself, where E_z is discontinuous and a pointwise comparison is ill-posed.

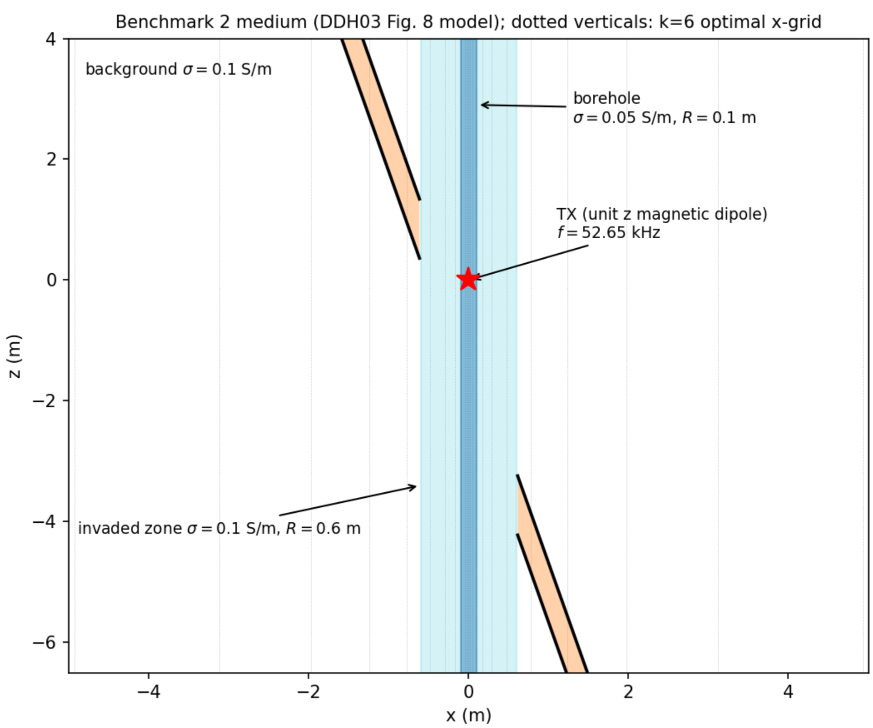

Figure 2: Benchmark 2 medium, x–z section (DDH03 Fig. 8 model). The invaded zone carries the background conductivity and replaces the anisotropic layer for $r < 0.6$ m, so the layer (orange) is an annular structure. Dotted verticals show the $k = 6$ optimal transverse grid.

Method. Full DDH03 methodology as in §1: optimal transverse grids, hybrid axial grid with the source exactly on a grid node, four-cluster sources and mixed boundary conditions, exact-geometry sub-cell homogenization (exact interface normals and volume fractions from the analytic geometry) by either the DDH03 eq.-(8/9) laminate ("Backus") scheme or the nodal scheme of [2], a *single fully coupled* sparse iterative solve (the off-diagonal conductivity couples the four clusters; solving them separately under-counts the anisotropy-generated cross-components), and interpolated four-cluster averaging at the exact receiver positions. Three refinement levels are run, scaling the transverse minimal step and the axial spacing together while growing k to hold the domain reach: L1 ($h_{\min} = 0.05$ m, $k = 6$, ≈ 121k unknowns — the published-comparison grid), L2 ($h_{\min} = 1/30$ m, $k = 7$, ≈ 223k), and L3 ($h_{\min} = 0.025$ m, $k = 8$, ≈ 368k). All iterative solves converged to 10^{-8} relative residual.

Absolute calibration of the reference. The published "no layer" curve serves double duty: comparing it to the closed-form homogeneous-medium solution shows agreement within 1–8% (the deficit appearing near the source, consistent with the borehole's local reduction of the field). This confirms that the published figure is in absolute SI units per unit dipole moment, so the comparison below involves no free scale.

Results. Figure 3 overlays the computed curves at the finest grid (lines) on the values digitized from the published figure (symbols).

- **No layer:** agreement within 0.3–4.5% at every receiver (geometric-mean ratio 1.021 Backus, 1.019 nodal at L3; log-scatter 1.1%), improving steadily with refinement (1.035 at L1). This confirms the absolute normalization and sets the accuracy floor of the digitization itself at the 1–2% level.
- **Resistive layer:** agreement within 3–12% (geometric-mean ratio 1.070 Backus, 1.077 nodal at L3; log-scatter 2%), and *flat across all three refinement levels*: the Backus ratio moves only $1.085 \to 1.073 \to 1.070$ from L1 to L3, and Backus and nodal agree with each other to better than 1.2% at every receiver. In the grid-calibration-free attenuation measure (geometric mean of the layer/no-layer ratio over the receivers) the published figure gives 0.697 and the averaged schemes give 0.729–0.742 at every level. The residual few percent is consistent with digitization error: both the layer's axis intercepts ($-0.95/-1.93$ m) and the published values themselves are read off the printed figure.

Resolution independence. The refinement behavior above is itself a validation of the DDH03 sub-cell averaging approach: with either averaged scheme the computed curves are essentially identical at all three grid levels, reproducing DDH03's own observation that their $k = 6$ and $k = 12$ results "can hardly be distinguished" even though the layer is thinner than the local grid cells. This is the frequency-domain counterpart of the thin-resistive-structure results of [2]: the sub-cell layer is captured by homogenization, not by resolution.

4 Remarks

The two benchmarks are complementary: Benchmark 1 has an exact external reference and tests the sharp-interface machinery and cluster averaging; Benchmark 2 tests the full anisotropic pipeline

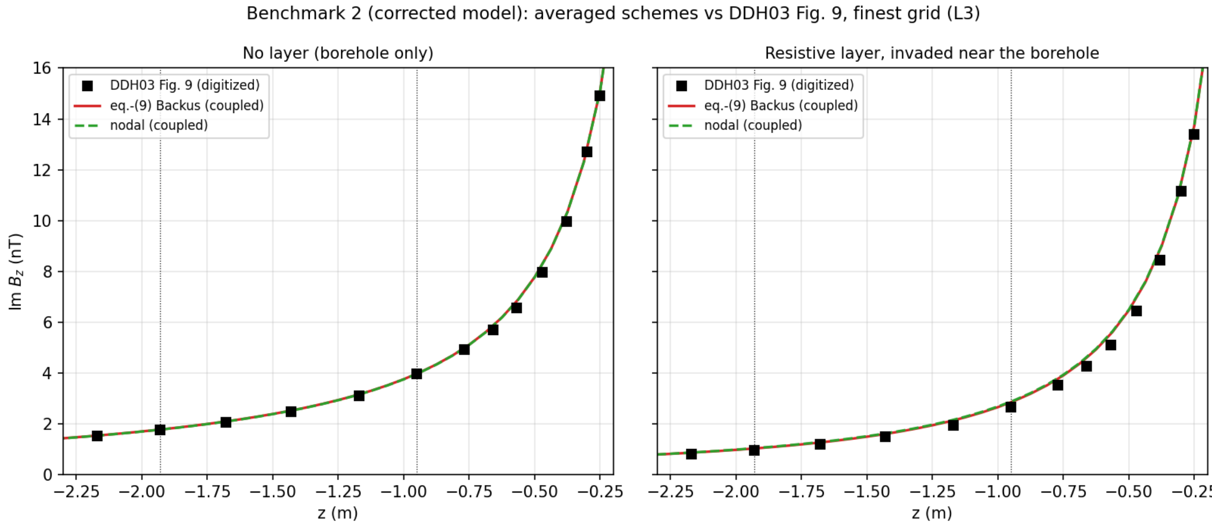

Figure 3: Benchmark 2: Im B_z on the axis at the finest grid (L3). Lines: `lebedev_em`, coupled solve with eq.-(9) Backus (solid red) and nodal (dashed green) averaging — the two are indistinguishable at this scale. Symbols: values digitized from DDH03 Fig. 9. Unit dipole moment on both sides; no fitted constants. Dotted verticals mark where the layer crosses the axis.

— tilted tensors, a borehole crossing a thin high-contrast layer, homogenization, the coupled four-cluster solve — against independently published curves whose absolute normalization we verified analytically, and its refinement study additionally validates the resolution-independence that is the central claim of the DDH03 averaging approach. Both succeed with unit dipole moment and no adjustable constants.

Work in progress: the deviated-borehole model of DDH03 Figs. 6–7 (borehole, invaded zone, and a 60° dipping anisotropic half-space) is being finalized and is omitted from these notes.

References

[1] S. Davydycheva, V. Druskin, T. Habashy, *An efficient finite-difference scheme for electromagnetic logging in 3D anisotropic inhomogeneous media*, Geophysics **68**(5):1525–1536, 2003.

[2] S. Moskow, V. Druskin, T. Habashy, P. Lee, S. Davydycheva, *A finite difference scheme for elliptic equations with rough coefficients using a Cartesian grid nonconforming to interfaces*, SIAM J. Numer. Anal. **36**:442–464, 1999.